Complete Diet & Nutrition Guide

Common health conditions, vitamins and minerals in one reference

Written around Indian foods, meal patterns and kitchen habits

This guide covers
Part A — Common conditions: type 2 diabetes, high blood pressure, thyroid disorders, high cholesterol, fatty liver, high uric acid and gout, PCOS, low haemoglobin and anaemia, and a note on kidney disease.
Part B — Vitamins: A, B1, B2, B3, B6, B9 (folate), B12, C, D, E and K.
Part C — Minerals: iron, calcium, zinc, magnesium, iodine and potassium.
Part D — Everyday practice: a universal healthy plate, food and medicine interactions, label reading, and the warning signs that need a doctor rather than a diet change.

Please read this first

- This is **general nutrition education**. It is not a personalised prescription, a diagnosis, or a treatment plan.
- Your own requirements change with your medication, kidney and liver function, pregnancy, age, allergies and other conditions. Advice that helps one condition can harm another — the potassium-rich foods recommended for blood pressure, for example, are restricted in kidney disease.
- Never start, stop or change a prescribed medicine on the basis of a diet guide.
- If you have a diagnosed condition, ask your doctor to refer you to a registered dietitian for a plan built around you.

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PART A — COMMON HEALTH CONDITIONS

Eating patterns for the conditions most often diagnosed on routine blood tests.

A1. Type 2 Diabetes and Prediabetes

The goal is steadier glucose through the day, a healthier weight, and lower long-term risk to the heart, kidneys, eyes and nerves. Diet works alongside medication, activity and sleep.

Core principles

- **Total carbohydrate matters most** — the amount in a meal drives the glucose rise more than the type, though type still counts.
- **Choose slow carbohydrates:** whole, minimally processed grains raise glucose more gently than refined flour.
- **Never eat carbohydrate alone.** Pair it with protein, fibre or healthy fat to blunt the spike.
- **Keep meal timing regular.** Skipping meals then over-eating causes large swings.
- **Aim for 25-30 g fibre a day** and, if overweight, a 5-10% weight loss — which improves control substantially.

The plate method

Divide a 9-inch plate: **half** non-starchy vegetables, **one quarter** protein, **one quarter** carbohydrate. In a thali this means more sabzi and dal, with a fixed roti or rice portion rather than an open-ended one.

Group	Choose freely	Limit or avoid
Grains	Jowar, bajra, ragi, whole wheat, oats, barley, brown rice, quinoa	Maida, white bread, naan, biscuits, cakes, instant noodles, large white-rice portions
Protein	Dal, chana, rajma, lobia, soya, paneer, curd, eggs, fish, skinless chicken	Deep-fried and processed meats, sausages, salami
Vegetables	Spinach, methi, lauki, tinda, bhindi, cabbage, cauliflower, capsicum, tomato, beans	Large portions of potato, sweet potato, yam, arbi — count these as carbohydrate
Fruit	Guava, apple, pear, papaya, orange, berries, jamun — whole, one serving at a time	Fruit juice, large amounts of mango/chikoo/banana, dried fruit, tinned fruit in syrup
Fats	Mustard, groundnut, rice bran, olive oil; a small handful of nuts and seeds	Vanaspati, reused frying oil, excess ghee and butter, fried snacks
Drinks	Water, unsweetened tea or coffee, buttermilk, lemon water without sugar	Soft drinks, packaged juice, sweetened lassi, energy drinks

Sugar and sweeteners

Jaggery, honey, brown sugar and khand raise blood glucose in essentially the same way as white sugar. Treating them as 'safe' is one of the most common and costly mistakes.

Safety notes

• If you take **insulin or sulfonylureas** (glimepiride, gliclazide, glibenclamide), cutting carbohydrate or fasting without adjusting the dose can cause dangerous hypoglycaemia. Speak to your doctor first, and carry a fast-acting sugar source.

• Kidney disease changes protein, potassium, phosphorus and salt targets — do not follow a high-protein plan without a renal dietitian.

• Alcohol can cause delayed low glucose, especially on an empty stomach.

A2. High Blood Pressure (Hypertension)

The DASH pattern — rich in vegetables, fruit, pulses and low-fat dairy, low in salt and refined food — is the best studied dietary approach and can lower systolic pressure meaningfully within weeks.

Targets

- **Salt under 5 g a day** (about 2,000 mg sodium — roughly one level teaspoon from all sources combined).
- **4-5 servings each** of vegetables and fruit daily.
- Every **1 kg of weight lost** tends to lower systolic pressure by about 1 mmHg.
- 150 minutes a week of brisk walking or equivalent; minimise alcohol.

Where the salt actually hides

High-salt items to cut back	Lower-salt swap
Pickles, papad, chutney powders, salted masalas	Fresh chutney without salt, lemon juice, amchur
Namkeen, chips, salted biscuits, bhujia	Unsalted roasted chana, makhana, nuts, fruit
Instant noodles, soup cubes, ready-to-eat mixes	Home-made soup, khichdi, poha
Sauces — soy, ketchup, chilli, salad dressings	Vinegar, lemon, garlic, herbs, black pepper
Processed meat, cheese spreads, tinned food	Fresh chicken or fish, paneer, fresh vegetables
Bakery bread, buns, rusk, baking soda in cooking	Whole wheat roti, home-made bread without added salt

Foods that help

- **Potassium-rich foods** counter sodium: banana, orange, musambi, coconut water, spinach, drumstick, beans, sweet potato, rajma, curd. *See the caution below if you have kidney disease.*
- **Magnesium and calcium:** low-fat milk and curd, ragi, sesame, leafy greens, almonds.
- **Healthy fats:** fatty fish twice a week, walnuts, flaxseed, mustard or olive oil.
- Beetroot and leafy greens supply dietary nitrate, which has a modest pressure-lowering effect.

Retraining your taste

Taste adapts in four to six weeks. Reduce gradually — about a quarter less salt each fortnight — build flavour with garlic, ginger, curry leaves, roasted cumin, kokum and lemon, and take the salt shaker off the table. On labels, above 600 mg sodium per 100 g is high; below 120 mg is low.

Safety notes

• **Do not use potassium-based salt substitutes** without medical advice if you take an ACE inhibitor, ARB or spironolactone, or have kidney disease — potassium can rise to dangerous levels.

• In chronic kidney disease, potassium-rich foods may need restricting, which reverses much of the advice above.

• Mulethi (licorice) in regular amounts raises blood pressure, as do many over-the-counter painkillers (NSAIDs) and decongestants.

A3. Thyroid Disorders

Diet supports thyroid health and helps with symptoms and weight, but it cannot correct a hormone deficiency or excess on its own — medication does that.

Nutrient	Why it matters	Sources
Iodine	Raw material for thyroid hormone; too little and too much both cause problems	Iodised salt (main source), fish, prawns, milk, curd, eggs
Selenium	Converts T4 into active T3 and protects the gland	Eggs, fish, sunflower seeds, mushrooms; 1-2 Brazil nuts daily is plenty
Zinc	Supports hormone production	Pulses, chana, pumpkin seeds, cashews, meat, seafood
Iron	Low iron blunts response to treatment and often coexists	Leafy greens, dates, jaggery, meat, fortified cereals

Nutrient	Why it matters	Sources
Vitamin D	Frequently low in autoimmune thyroid disease	Sunlight, fortified milk, eggs, fatty fish

Underactive thyroid (hypothyroidism)

- Keep **iodised salt** as your regular salt, within the same 5 g daily limit that applies to everyone.
- Build meals around **protein** — it protects muscle and helps satiety when metabolism has slowed.
- **Fibre and fluids** help the constipation that often accompanies it.
- Weight gain responds to modest calorie reduction plus resistance exercise, not crash dieting.
- There is **no evidence a gluten-free diet helps** Hashimoto's unless you also have coeliac disease.

What about goitrogens?

Cabbage, cauliflower, broccoli, soya and mustard greens contain compounds that can interfere with iodine uptake, but this only matters with very large raw amounts over long periods, and cooking deactivates most of the effect. **Normal cooked portions are fine and should not be avoided.** Soya is the genuine concern, because it reduces absorption of thyroid medication.

Overactive thyroid (hyperthyroidism)

- Energy and protein needs rise — eat regular, adequate meals rather than restricting.
- **Avoid iodine supplements, kelp and seaweed**, which can worsen the condition.
- Cut caffeine, which amplifies palpitations, tremor and poor sleep; protect bone with calcium and vitamin D.

Medication timing — more important than any food choice

- Take **levothyroxine on an empty stomach**, 30-60 minutes before breakfast, with plain water.
- Leave **at least 4 hours** between levothyroxine and calcium supplements, iron supplements, antacids and multivitamins.
- Soya, high-fibre meals and coffee too close to the tablet also reduce absorption — allow 30-60 minutes.
- Consistency matters more than the exact time of day. Dose changes are guided by blood tests, never by how you feel.

A4. High Cholesterol and Heart Health

- **Replace, don't just remove.** Swapping saturated fat for unsaturated fat lowers LDL; swapping it for refined carbohydrate does not.
- **Cut saturated fat:** vanaspati, excess ghee and butter, coconut oil in quantity, fatty red meat, full-fat cream, bakery items, fried snacks.
- **Avoid trans fat entirely** — 'partially hydrogenated' on a label, and reused frying oil.
- **Add soluble fibre:** oats, barley, methi seeds, isabgol, rajma, chana, apple, guava — 5-10 g a day lowers LDL measurably.
- **Add unsaturated fat:** mustard, rice bran, groundnut and olive oil, walnuts, almonds, flaxseed, fatty fish twice a week.
- Dietary cholesterol from eggs matters far less than saturated fat does for most people; one egg a day is reasonable.
- Alcohol and sugar mainly drive **triglycerides** — if those are high, cut both first.

A5. Fatty Liver (NAFLD)

- **Weight loss is the single most effective treatment.** Losing 7-10% of body weight can reverse liver inflammation.
- **Cut fructose hardest:** soft drinks, packaged juice, sweetened beverages and sweets are the strongest dietary drivers.
- Reduce refined carbohydrate: maida, white rice in quantity, bakery, sugary snacks.
- Favour a Mediterranean-style pattern: vegetables, pulses, whole grains, olive or mustard oil, nuts, fish.
- **Stop alcohol.** Even moderate intake accelerates damage in a fatty liver.
- Coffee (unsweetened) is consistently associated with less liver fibrosis.

- Aim for 150-200 minutes of activity a week — exercise reduces liver fat even without weight loss.

A6. High Uric Acid and Gout

Limit	Choose freely
Organ meats (liver, kidney, brain) — the highest purine foods	Vegetables, including the ones often wrongly banned
Red meat, mutton, and shellfish/prawns	Low-fat milk, curd and paneer — these actively lower uric acid
Beer and spirits — among the strongest triggers	Cherries and vitamin C-rich fruit
Sugary drinks and anything with high-fructose corn syrup	Whole grains, eggs, coffee, plenty of water (2.5-3 litres)
Yeast extracts and meat gravies	Nuts and pulses in normal amounts

A common myth worth correcting

• Tomato, spinach, cauliflower, mushroom, peas and dal are often eliminated unnecessarily. **Plant purines do not raise gout risk the way animal purines and alcohol do**, and the evidence does not support cutting these nutritious foods out.

• Diet influences uric acid modestly. If you have recurrent gout attacks, urate-lowering medication does far more than diet alone.

A7. PCOS (Polycystic Ovary Syndrome)

- Insulin resistance underlies most PCOS, so the **diabetes principles in A1 apply directly** — slow carbohydrates, protein at every meal, regular timing.
- A modest **5-10% weight loss** can restore ovulation and regular cycles where weight is a factor.
- Prioritise protein and fibre at breakfast; skipping it worsens cravings and afternoon crashes.
- Check for vitamin D and B12 deficiency — both are common, and metformin lowers B12 over time.
- Resistance training improves insulin sensitivity more reliably than cardio alone.
- No single food causes or cures PCOS. Be sceptical of elimination diets sold online.

A8. Low Haemoglobin — Foods That Raise It

Haemoglobin is the protein in red blood cells that carries oxygen. When it falls, you feel tired, breathless on stairs, dizzy or cold, and look pale — particularly in the lower eyelids, tongue and nail beds. This section covers what actually raises it, and which widely-believed foods do far less than their reputation suggests.

Typical reference ranges

Group	Usual normal range (g/dL)	Commonly called low below
Adult men	13.0 – 17.0	13.0
Adult women (non-pregnant)	12.0 – 15.0	12.0
Pregnant women	11.0 – 14.0	11.0
Children (6–12 years)	11.5 – 15.5	11.5

Ranges differ slightly between laboratories — always read the range printed on your own report, and interpret it alongside MCV, MCH and ferritin rather than alone.

Before you start eating for it — two things to understand

• **Not all low haemoglobin is iron deficiency.** It can also come from vitamin B12 or folate deficiency, thalassaemia or other inherited conditions, chronic kidney disease, long-term inflammation, or blood loss. Iron-rich food will not help if iron is not the problem, and iron supplements taken unnecessarily can cause harm.

• **Diet alone works slowly.** It is excellent for prevention and maintenance, but an established deficiency usually needs a prescribed supplement. Expect roughly **1 g/dL rise every 2–4 weeks** with proper treatment — diet on its own is considerably slower.

• Low haemoglobin is a finding, not a diagnosis. The **cause** must be identified — heavy periods, gut bleeding, ulcers, piles, coeliac disease, hookworm or poor intake. Unexplained anaemia in men and post-menopausal women always needs investigation.

The strongest haemoglobin-raising foods

Values below are approximate iron content per 100 g of the raw food. Published Indian food tables vary, and newer analyses give lower figures than older ones for leafy vegetables, so treat these as a ranking rather than exact numbers.

Food	Iron (mg/100 g, approx.)	How to use it
Garden cress seeds (halim / aliv)	Very high — among the richest	1 tsp soaked in milk or water, or as ladoo; a traditional post-partum food for good reason
Liver (mutton or chicken)	6 – 10	The single most effective food source; well absorbed. Once a week is plenty
Sesame seeds (til), black or white	9 – 14	Til ladoo, chutney powder, sprinkled on sabzi
Soyabean and soya chunks	8 – 11	Curry, pulao, or added to mince dishes
Pumpkin seeds	8 – 9	A handful as an evening snack
Bajra (pearl millet)	6 – 8	Replace wheat roti at one meal a day
Amaranth / rajgira, and its leaves	8 – 9	Rajgira roti, laddoo, or the leaves as sabzi
Whole chana, rajma, lobia, masoor	5 – 9	Daily dal or a bean sabzi; soak and sprout for better absorption
Ragi (finger millet)	4 – 5	Ragi porridge, roti or dosa — also rich in calcium
Dried dates, anjeer, raisins, apricots	2 – 4	A few daily; useful but not a substitute for the above
Jaggery	2 – 11 (varies widely)	Modest amounts. **Not suitable in diabetes**
Mutton and red meat	2 – 3	Well-absorbed haem iron; also boosts absorption of plant iron in the same meal
Prawns and fish	2 – 3	Twice a week
Eggs	1 – 2	Modest content, moderately absorbed — useful as part of the pattern
Poha (flattened rice)	Often fortified — check the pack	Breakfast with peanuts and lemon

Popular 'blood-building' foods that do less than you think

• **Beetroot** — low in iron. Its reputation comes from the colour. It does supply folate, which is genuinely useful, but it will not raise haemoglobin by itself.

• **Pomegranate and apple** — both are low in iron. Pomegranate's vitamin C helps absorb iron from *other* foods, which is the real benefit.

• **Spinach** — still worth eating, but older Indian tables badly overstated its iron (early analyses were contaminated by soil and iron utensils). Its oxalates also reduce absorption. Bajra, sesame and pulses beat it comfortably.

• **Milk** — nutritious, but low in iron and it actively blocks iron absorption. Large milk intakes are a recognised cause of anaemia in toddlers.

Absorption matters as much as intake

Plant (non-haem) iron is only 2–10% absorbed, against 15–35% for animal (haem) iron. Changing *what you eat alongside it* can easily double what your body actually takes in.

Do this — helpers	Avoid this — blockers
Add vitamin C to every iron-rich meal — lemon over dal and sabzi, amla, guava, orange, tomato, capsicum. The strongest enhancer there is	**Tea and coffee with or just after meals** — tannins can cut absorption by more than half. Keep a 1–2 hour gap
Soak, sprout and ferment — idli, dosa, dhokla, sprouted moong and chana release far more iron than the unsoaked versions	**Milk, curd, paneer and calcium supplements** taken at the same time as iron-rich meals or iron tablets
Cook acidic dishes in an iron kadhai — tomato or tamarind-based curries pick up a useful amount	**Antacids and acid-reducing medicines (PPIs)** — stomach acid is needed to absorb iron
Include a little meat, fish or egg with a plant-iron meal — it improves absorption of the whole meal	**Excess bran, wheat husk and unsoaked whole grains** — phytates bind iron

Don't forget B12 and folate

If your MCV is **high** rather than low, the problem is more likely vitamin B12 or folate than iron — and iron-rich food will not correct it. Include milk, curd, paneer, eggs and fish for B12 (see section B7), and dark leafy greens, methi, chana and orange for folate (section B6). Long-term vegetarians and people on metformin are the most likely to need B12.

A sample day built around raising haemoglobin

Time	What to eat	Why it works
Early morning	Iron tablet with water or orange juice, if prescribed. Soaked halim seeds in water	Best absorbed on an empty stomach, away from milk and tea
Breakfast	Poha with peanuts, curry leaves and a good squeeze of lemon; or besan chilla with mint chutney	Fortified poha plus vitamin C from lemon
Mid-morning	1 guava, orange, or amla in any form	High vitamin C between meals
Lunch	Bajra roti, palak-chana dal, sprouted moong salad with tomato and lemon, curd on the side	Millet and pulse iron, sprouting plus vitamin C. Keep the curd small at this meal
Evening	Roasted chana, or a til-and-jaggery ladoo, or a few dates with pumpkin seeds	Dense iron in snack form. Have tea at least an hour away from this
Dinner	Chicken, fish or rajma curry with rice and a mixed vegetable	Haem iron lifts absorption from everything else on the plate
Through the day	Tea and coffee only between meals, never with them	The single highest-impact habit change

Realistic expectations and safety

• Give it time. Haemoglobin rises gradually — **recheck after 8–12 weeks**, not after two weeks.

• If you are prescribed iron, treatment usually continues **3–6 months after haemoglobin normalises** to refill stores. Stopping as soon as you feel better is the commonest reason people relapse.

• Dark stools are expected and harmless. Constipation and nausea are common; taking the tablet with food eases this but lowers absorption somewhat. **Alternate-day dosing** may absorb better — ask your doctor.

• **Never take iron without a confirmed deficiency.** Iron overload damages the liver and heart, and iron tablets are a leading cause of poisoning in young children — store them out of reach.

• See a doctor promptly if haemoglobin is very low, if you have chest pain or breathlessness at rest, black or bloody stools, or if levels do not improve on treatment.

A9. A Note on Chronic Kidney Disease

Kidney disease overrides the rest of this guide

• In CKD, **potassium, phosphorus, protein, salt and fluid** may all need restriction, and the limits depend on your stage and blood results.

• This reverses much of the standard advice: the bananas, coconut water, spinach, rajma, nuts, dal and low-sodium salt substitutes recommended elsewhere here may be unsafe for you.

• Do not self-manage. **A renal dietitian's individual plan replaces every other section of this document.**

PART B — VITAMINS

What each vitamin does, how deficiency shows up, and where to find it in everyday food.

A varied diet covers most of these. The ones Indian adults most often fall short on are **vitamin D, vitamin B12** and, in women, **iron and folate** — these have fuller sections. Deficiency should be confirmed by a blood test, not guessed from symptoms, because tiredness and hair fall are common to almost every entry below.

B1. Vitamin A

What it does	Signs it may be low	Best food sources
Vision (especially in dim light), immunity, skin and the lining of the gut and lungs.	Night blindness, dry eyes, rough or dry skin, frequent infections.	Carrot, pumpkin, sweet potato, papaya, mango, spinach, methi, drumstick leaves, egg yolk, milk, liver, fortified oils.

Note

• It is fat-soluble — eat these with a little oil or ghee for absorption. High-dose supplements can be toxic and are **unsafe in pregnancy**; food sources are not.

B2. Vitamin B1 (Thiamine)

What it does	Signs it may be low	Best food sources
Converts carbohydrate into energy; essential for nerve and heart function.	Tingling or burning feet, leg weakness, poor appetite, irritability, swelling.	Whole grains, hand-pounded or unpolished rice, wheat germ, pulses, groundnut, sunflower seeds, pork, eggs.

Note

• Deficiency is much more likely with **heavy alcohol use**, prolonged vomiting or a diet based on polished white rice. Alcohol-related thiamine deficiency is a medical emergency, not a diet issue.

B3. Vitamin B2 (Riboflavin)

What it does	Signs it may be low	Best food sources
Energy release from food; healthy skin, eyes and mucous membranes.	Cracks at the corners of the mouth, sore or magenta-coloured tongue, gritty eyes, scaly skin around the nose.	Milk, curd, paneer, eggs, green leafy vegetables, almonds, mushrooms, whole grains, liver.

Note

• Riboflavin is destroyed by light — milk stored in clear containers in sunlight loses a significant amount.

B4. Vitamin B3 (Niacin)

What it does	Signs it may be low	Best food sources
Energy metabolism, skin health, nervous system and digestive function.	Rough pigmented skin on sun-exposed areas, diarrhoea, mouth soreness, confusion in severe cases.	Groundnut, whole grains, pulses, chicken, fish, eggs, mushrooms, sunflower seeds.

Note

• The body can make niacin from the amino acid tryptophan, so an adequate protein intake protects against deficiency. Deficiency is classically linked to maize-dominant diets.

B5. Vitamin B6 (Pyridoxine)

What it does	Signs it may be low	Best food sources
Protein metabolism, haemoglobin formation, immune function and brain chemistry.	Cracked lips, sore tongue, low mood, confusion, tingling hands and feet, a specific type of anaemia.	Chana, rajma, banana, potato, chicken, fish, sunflower seeds, pistachios, fortified cereals.

Note

• **Isoniazid** (TB treatment) depletes B6, which is why pyridoxine is routinely prescribed alongside it. Very high self-prescribed doses over long periods can themselves cause nerve damage.

B6. Vitamin B9 (Folate / Folic Acid)

What it does	Signs it may be low	Best food sources
Cell division, red blood cell formation, and normal development of the baby's spine and brain in early pregnancy.	Large-cell anaemia, tiredness, mouth ulcers, low mood, poor appetite.	Dark leafy greens, methi, spinach, broccoli, beetroot, chana, rajma, lobia, orange, banana, groundnut, fortified cereals.

Pregnancy and a caution

• **Folic acid supplementation should start before conception** and continue through the first trimester — it substantially reduces neural tube defects. Waiting until pregnancy is confirmed is often too late.

• Folate supplements can correct the blood picture of **B12 deficiency while nerve damage continues underneath**. The two must always be assessed together.

Note

• Folate is heat-sensitive — heavy boiling destroys much of it; steam or stir-fry instead.

B7. Vitamin B12 (Cobalamin)

What it does	Signs it may be low	Best food sources
Red blood cell formation, nerve function and DNA synthesis.	Tiredness, pallor, tingling or numb hands and feet, burning feet, balance problems, memory or mood changes, sore tongue.	**Animal foods only:** milk, curd, paneer, cheese, eggs, fish, prawns, chicken, mutton, liver. For vegans: fortified soya or oat milk, fortified cereals, fortified nutritional yeast.

At higher risk	Why
Long-term vegetarians	Dairy alone often supplies too little
Vegans	Essentially no natural source without fortified foods
People on long-term metformin	Reduces B12 absorption over years
People on long-term antacids or PPIs	Stomach acid is needed to release B12 from food
Older adults; after stomach or bowel surgery	Absorption declines with age and with altered anatomy

Be realistic about diet here

• A vegetarian diet supplies some B12 through dairy, but usually not enough to correct a real deficiency. **Established deficiency needs supplements or injections**, not just dietary change.

• Neurological symptoms need prompt treatment — prolonged deficiency can cause **lasting nerve damage**.

• Lifelong vegetarians and vegans are sensible to take a regular maintenance supplement even once levels are normal. Discuss the dose with your doctor.

B8. Vitamin C (Ascorbic Acid)

What it does	Signs it may be low	Best food sources
Collagen for skin, gums and blood vessels; wound healing; immunity; and it dramatically improves iron absorption.	Bleeding or swollen gums, easy bruising, slow wound healing, rough bumpy skin, joint aches.	Amla (exceptionally rich), guava, orange, musambi, lemon, capsicum, tomato, cabbage, broccoli, drumstick leaves, sprouts.

Note

• Destroyed by prolonged cooking and by cutting long before eating — prefer raw or lightly cooked. Pairing a vitamin C food with an iron-rich meal is one of the highest-value habits in this whole guide.

B9. Vitamin D

What it does	Signs it may be low	Best food sources
Calcium absorption, bone strength, muscle function and immune regulation.	Bone and lower back pain, muscle weakness, difficulty climbing stairs or rising from squatting, frequent infections, low mood.	Sunlight is the main source. Food: fatty fish (bangda, sardine, rohu, salmon), egg yolk, fortified milk and cereals, UV-exposed mushrooms.

Sunlight — the practical detail

- Roughly **15-30 minutes of midday sun** (about 10 am to 3 pm) on arms, legs or back, several times a week.
- Darker skin needs considerably longer to make the same amount.
- **Glass blocks the relevant UVB**, so sitting by a window does not work.
- Avoid burning — balance this against skin cancer risk; hours of exposure add nothing useful.

Getting the most from a supplement

- Vitamin D is fat-soluble — take it with a meal containing some fat.
- Ensure adequate calcium (milk, curd, paneer, ragi, sesame, leafy greens) and magnesium, which is needed to activate it.
- Deficiency is extremely common in India despite abundant sunshine, because of indoor lifestyles, covered clothing, pollution and darker skin. **Very few people reach adequacy from food alone.**

Safety note

• **Vitamin D is fat-soluble and accumulates.** Very high doses over long periods raise blood calcium and can damage the kidneys. Follow the prescribed schedule, retest as advised, and do not continue high-dose supplements indefinitely by default.

B10. Vitamin E

What it does	Signs it may be low	Best food sources
Antioxidant protection for cell membranes; skin and immune health.	Rare from diet alone. Seen mainly in fat-malabsorption conditions: muscle weakness, poor coordination, numbness.	Sunflower and safflower oil, wheat germ, almonds, sunflower seeds, groundnut, spinach, avocado.

Note

• Dietary vitamin E is beneficial, but **high-dose supplements have not shown benefit** and at large doses may increase bleeding risk, especially with blood thinners.

B11. Vitamin K

What it does	Signs it may be low	Best food sources
Blood clotting and bone metabolism.	Easy bruising, bleeding gums, slow clotting. Deficiency is uncommon in adults eating greens.	Palak, methi, mustard greens, cabbage, broccoli, coriander, soybean oil; vitamin K2 from curd, cheese and fermented foods.

Note

• **If you take warfarin, do not suddenly increase or decrease green leafy vegetables** — vitamin K directly opposes the drug. Consistency from day to day matters far more than the amount. Newer anticoagulants are not affected.

PART C — MINERALS

The minerals most often found low on Indian blood reports, and the ones easiest to get wrong.

C1. Iron

The most common nutritional deficiency worldwide. Absorption matters as much as intake, because plant iron is poorly absorbed on its own.

Haem iron — well absorbed (15-35%)	Non-haem iron — poorly absorbed (2-10%)
Liver, mutton, red meat	Bajra, ragi, jowar, amaranth (rajgira)
Chicken, especially dark meat	Dal, chana, rajma, lobia, soyabean, sprouts
Fish and prawns	Spinach, methi, mustard greens, drumstick leaves
Eggs (modest, less well absorbed)	Sesame and pumpkin seeds, garden cress (halim)
	Dates, raisins, anjeer, jaggery, poha, fortified cereals

Helpers — pair with iron meals	Blockers — keep 1-2 hours apart
Vitamin C: amla, lemon, guava, orange, tomato, capsicum	Tea and coffee (tannins) — the biggest practical blocker
Soaking, sprouting and fermenting (idli, dosa, dhokla)	Milk, curd, paneer and calcium supplements with iron tablets
Cooking acidic food in an iron kadhai	Antacids and acid-reducing medicines (PPIs)
A little meat or fish alongside plant iron	Very large amounts of bran and wheat husk

Supplement and follow-up notes

- Dark stools are expected and harmless. Nausea and constipation are common; taking the tablet with food reduces side effects but also reduces absorption.
- **Alternate-day dosing** may be absorbed better than daily dosing for some people — ask your doctor.
- Treatment usually continues **3-6 months after haemoglobin normalises** to refill stores. Stopping as soon as you feel better leads to relapse. Recheck haemoglobin and ferritin at 8-12 weeks.
- Never take iron without a confirmed deficiency. **Iron overload is harmful**, and iron tablets are a leading cause of poisoning in young children — store them out of reach.

C2. Calcium

What it does	Signs it may be low	Best food sources
Bone and tooth structure, muscle contraction, nerve signalling and blood clotting.	Muscle cramps, numbness or tingling, brittle nails; long term, low bone density and fractures.	Milk, curd, paneer, cheese, ragi (exceptionally rich), sesame (til), almonds, drumstick and amaranth leaves, small fish eaten with bones, tofu, fortified plant milks.

Note

- Calcium is useless without adequate **vitamin D**, so correct both together. Split large doses — absorption is better at 500 mg at a time. Keep calcium supplements at least 4 hours from thyroid medication and 2 hours from iron.

C3. Zinc

What it does	Signs it may be low	Best food sources
Immunity, wound healing, taste and smell, skin repair, growth and male fertility.	Slow wound healing, frequent infections, hair fall, loss of taste, mouth ulcers, acne that does not settle.	Pumpkin seeds, sesame, cashew, chana, rajma, whole grains, curd, eggs, chicken, mutton, shellfish.

Note

• Phytates in unsoaked grains and pulses reduce absorption — soaking and sprouting help significantly, which matters most for vegetarians. Prolonged high-dose zinc supplementation can cause copper deficiency.

C4. Magnesium

What it does	Signs it may be low	Best food sources
Over 300 enzyme reactions, muscle and nerve function, blood pressure regulation, and activating vitamin D.	Muscle cramps and twitching, restless legs, poor sleep, fatigue, irritability, palpitations.	Pumpkin and sunflower seeds, almonds, cashew, whole grains, bajra, dal, rajma, spinach, banana, dark chocolate.

Note

• Commonly low in people with type 2 diabetes and in heavy alcohol use. If a vitamin D level will not rise despite supplementation, low magnesium is one reason to consider.

C5. Iodine

What it does	Signs it may be low	Best food sources
The raw material for thyroid hormone, and essential for brain development in pregnancy and infancy.	Goitre (neck swelling), features of an underactive thyroid, and impaired development in children.	Iodised salt — the principal and intended source. Also fish, prawns, milk, curd and eggs.

Note

• Store iodised salt in a closed container away from heat and light; iodine is lost when it is left open. **Rock salt, sendha namak and most 'natural' salts contain little or no iodine.** Excess iodine from kelp or supplements is harmful and can trigger thyroid dysfunction — more is not better.

C6. Potassium

What it does	Signs it may be low	Best food sources
Fluid balance, nerve and muscle function, and counteracting the blood-pressure effect of sodium.	Muscle weakness and cramps, constipation, palpitations, fatigue.	Banana, orange, musambi, coconut water, potato and sweet potato, spinach, drumstick, rajma, dal, curd, dates.

Note

• **This is the single most condition-dependent entry in the guide.** Potassium-rich foods are actively encouraged for blood pressure but may be dangerous in chronic kidney disease, or when taking ACE inhibitors, ARBs or spironolactone. Follow your doctor's advice over the general recommendation.

PART D — EVERYDAY PRACTICE

The habits that apply regardless of which condition brought you to this guide.

D1. A Universal Healthy Plate

Almost every recommendation in Parts A to C is a variation on the same pattern. If you remember nothing else, remember this.

- **Half your plate vegetables**, in more than one colour, at both main meals.
- **Protein at every meal** — dal, chana, rajma, curd, paneer, eggs, fish or chicken. Most Indian diets are short on this.
- **Whole grains instead of refined** — jowar, bajra, ragi, whole wheat, unpolished rice.
- **Salt under 5 g a day**, and added sugar as low as you can keep it.
- **Fruit whole, not juiced** — juicing removes the fibre and concentrates the sugar.
- **Cook at home more often.** Restaurant and packaged food is where most excess salt, sugar and refined oil enters the diet.
- **Water as the default drink** — 2.5 to 3 litres a day unless your doctor has restricted fluids.
- **Move after meals.** Even a ten-minute walk blunts the glucose and triglyceride rise.

D2. A Sample Day You Can Adapt

Time	Meal	Adapt it for
Early morning	Warm water with lemon; 5 soaked almonds or 2 walnuts	Diabetes: no honey. BP: no salted nuts
Breakfast	Vegetable poha, besan chilla, or idli-sambar; plus 1 bowl curd	Diabetes: add protein. Anaemia: add lemon. BP: skip pickle and papad
Mid-morning	1 guava, orange or papaya	Anaemia: choose citrus or amla for the vitamin C
Lunch	2 jowar or wheat rotis, 1 bowl dal, large bowl sabzi, salad, curd	Diabetes: fixed roti count, eat salad first. CKD: follow your renal plan
Evening	Roasted chana or makhana; unsweetened tea or buttermilk	Anaemia: keep tea at least an hour from meals. BP: unsalted only
Dinner	1-2 rotis or small bowl rice, paneer/fish/dal, plenty of vegetables	Diabetes: finish 2-3 hours before bed. Fatty liver: no alcohol

D3. Food and Medicine Interactions Worth Knowing

Medicine	Interaction	What to do
Levothyroxine (thyroid)	Calcium, iron, antacids, soya, coffee and fibre all reduce absorption	Empty stomach, 30-60 min before food; 4-hour gap from calcium and iron
Warfarin	Vitamin K in leafy greens directly opposes the drug	Keep green intake **consistent** day to day rather than avoiding it
Metformin	Lowers vitamin B12 absorption over years	Have B12 checked periodically
ACE inhibitors, ARBs, spironolactone	Raise blood potassium	Avoid potassium-based salt substitutes; discuss high-potassium foods
Antacids and PPIs	Reduce absorption of B12, iron, calcium and magnesium	Review long-term use with your doctor; monitor levels
Statins	Grapefruit juice raises blood levels of some statins	Avoid grapefruit juice with these

Medicine	Interaction	What to do
Iron tablets	Tea, coffee, milk and calcium reduce absorption; vitamin C increases it	Take with water or citrus, away from tea and dairy
Isoniazid (TB)	Depletes vitamin B6	Pyridoxine is normally co-prescribed — take it

D4. Reading a Food Label

- Check the **per 100 g** column, not the per-serving one — serving sizes are often unrealistically small.
- **Sodium:** above 600 mg per 100 g is high; below 120 mg is low.
- **Sugar:** look beyond 'sugar' — glucose syrup, maltodextrin, invert syrup, fruit juice concentrate, honey and jaggery all count.
- **Fat:** 'partially hydrogenated' means trans fat. Avoid it.
- Ingredients are listed by weight — if sugar or maida is in the first three, the claim on the front matters less than what is in the list.
- 'Sugar-free' does not mean carbohydrate-free, and 'baked not fried' does not mean low in salt.

D5. When to See a Doctor Rather Than Change Your Diet

Seek medical advice promptly for any of these

• Chest pain, breathlessness at rest, fainting, or a fast irregular heartbeat.

• Blood in stool or urine, black tarry stools, or vomiting blood.

• Unexplained weight loss, persistent fever, or a lump anywhere.

• New tingling, numbness, burning feet, weakness or balance problems — these can indicate B12 deficiency causing nerve damage, which needs treatment without delay.

• Severe or persistent headache with very high blood pressure readings.

• Repeated low-glucose episodes if you are on diabetes medication.

• Swelling of the feet or face, or a marked drop in how much urine you pass.

• Any deficiency that does not improve after a completed course of treatment — the underlying cause has not been found.

D6. How to Use This Guide Well

- **Confirm before you correct.** Tiredness and hair fall fit almost every deficiency in Part B. Test, then treat.
- **Do not stack supplements.** Fat-soluble vitamins (A, D, E, K) and iron accumulate and can cause harm at high doses.
- **Change two or three habits at a time**, not fifteen. Adherence beats perfection.
- **Where two sections conflict, the more restrictive medical condition wins** — kidney disease over blood pressure, for instance. When in doubt, ask your doctor.